

From Phyllite To Landslides

Phyllite's Foliation As a Driver of Landslide Susceptibility in the Garhwal Himalaya

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Introduction

On the Dayara Bugyal trek in the Uttarkashi district of the Garhwal Himalaya, India, silvery, lustrous rocks were scattered across the ground along the trail (Figure 1). What stood out beyond their appearance was their layered structure and how easily they split into thin flakes. The trek leader mentioned that the area is highly prone to landslides, including the 2013 Kedarnath disaster roughly 80 km southeast, where heavy monsoon rainfall triggered over 6,000 landslides across Uttarakhand, killing more than 6,000 people (Martha et al., 2015; Allen et al., 2015). This paper explores the connection between that rock and the region's landslide susceptibility, tracing it from the rock's mineral chemistry to the physical processes that cause slopes in the Garhwal Himalaya to fail.



Figure 1: **Phyllite outcrop along the Dayara Bugyal trail, Uttarkashi district, Garhwal Himalaya.** The silvery luster and finely laminated surface visible in the field are characteristic of phyllite's foliated texture.

Rock and Its Chemistry

The rock was identified as phyllite by comparing trail samples (Figure 2) with geological references. The silvery luster, aphanitic texture in which individual mineral grains are invisible to the naked eye, and the wavy foliation that splits into parallel sheets match phyllite's defining characteristics as a low-grade foliated metamorphic rock sitting between slate and schist on the metamorphic spectrum (Figure 3).



Figure 2: **Hand samples of phyllite collected from the Dayara Bugyal trail.** The wavy foliation planes and fine-grained, aphanitic texture are visible across the samples, consistent with low-grade greenschist-facies metamorphism.

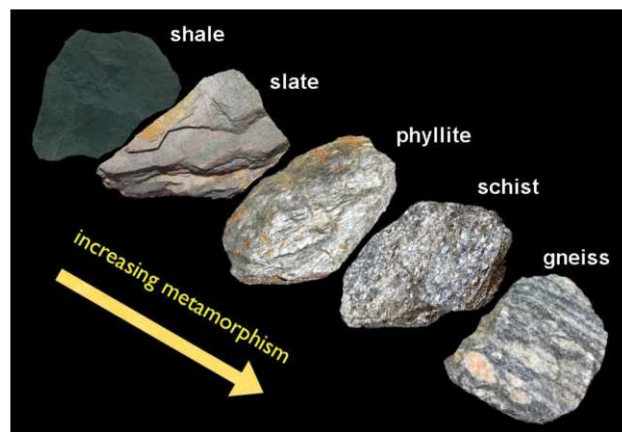


Figure 3: **The metamorphic rock sequence from shale to gneiss with increasing metamorphic grade.** Phyllite stands in the intermediate position between slate and schist, reflecting the moderate temperature and pressure conditions under which clay minerals recrystallize into platy phyllosilicates such as sericite and chlorite. Source: Gorokhovich, Y. (n.d.). *Rocks and rock cycle (II): Metamorphism and characteristics of metamorphic rocks.* Geology 101 for Lehman College (CUNY). <https://pressbooks.cuny.edu/gorokhovich/chapter/rocks-and-rock-cycle-ii-characteristics-of-metamorphic-rocks/>

The phyllite originated over a billion years ago, during the Proterozoic, when fine-grained marine sediments, predominantly shales rich in clay minerals, were deposited in a shallow-water basin at the margin of the ancient Indian craton (Mandal et al., 2021). These sediments were buried and incorporated into the growing Himalayan orogen as the Indian plate began its prolonged collision with Eurasia during the Cenozoic, producing regional metamorphism (Yin & Harrison, 2000). In the Garhwal Himalaya specifically, burial within the Lesser Himalayan sequence subjected these protoliths to temperatures and pressures sufficient to drive solid-state recrystallization without fully melting them (Oliver et al., 1995). Tectonic compression during this event forced new platy minerals to grow in parallel alignment rather than remaining randomly oriented as in the original shale, producing the fabric that gives phyllite its characteristic splitting behavior and weak directional strength.

At the mineral scale, the platy minerals produced during metamorphism are predominantly sericite, muscovite, and chlorite. Sericite’s crystal structure consists of T-O-T layers in which two tetrahedral silicate sheets are bonded to a central octahedral aluminium sheet, forming repeating sandwiches stacked on top of one another (Wang et al., 2019). The bonds within each sandwich are strong covalent and ionic bonds, but the bonds between sandwiches are weak, mediated by K^+ cations sitting in interlayer sites. This asymmetry explains why sericite has basal cleavage and a Mohs hardness of only 2 to 2.5, and why the rock as a whole has planes of weakness when billions of sericite grains align in the same direction during metamorphism. Phyllite’s foliation is sericite’s crystal cleavage scaled from angstroms to outcrop, which is the direct structural link between the mineral and the landslide susceptibility that Gerrard (1994) documented when he ranked phyllite as the most landslide-prone rock type in the Himalaya, ahead of shale, schist, sandstone, gneiss, granite, and quartzite.

Foliation to Failure

The foliation planes exposed along the trek dip roughly parallel to the slope surface (see Figure 4), turning phyllite’s mineral-scale weakness into slope-scale instability. At the mineral level, sericite interlayer bonds create cleavage planes with very low frictional resistance (Wang et al., 2019); at the rock level, metamorphic alignment stacks those planes into continuous foliation surfaces (Oliver et al., 1995); at the slope level, those surfaces become failure planes oriented where gravitational shear stress is highest. Karacan and Alca (2018) tested four foliated metamorphic rock types and found that phyllite shows the largest strength differences of all, with strength minimized when foliation is oriented between 30 and 60 degrees to the principal stress, which corresponds to typical slope angles in the Garhwal Himalaya (Rafique et al., 2024). No single level alone explains landslide susceptibility: without slope-parallel geometry, the mineral weakness is irrelevant, and without the mineral weakness, the geometry produces no preferential failure surface.

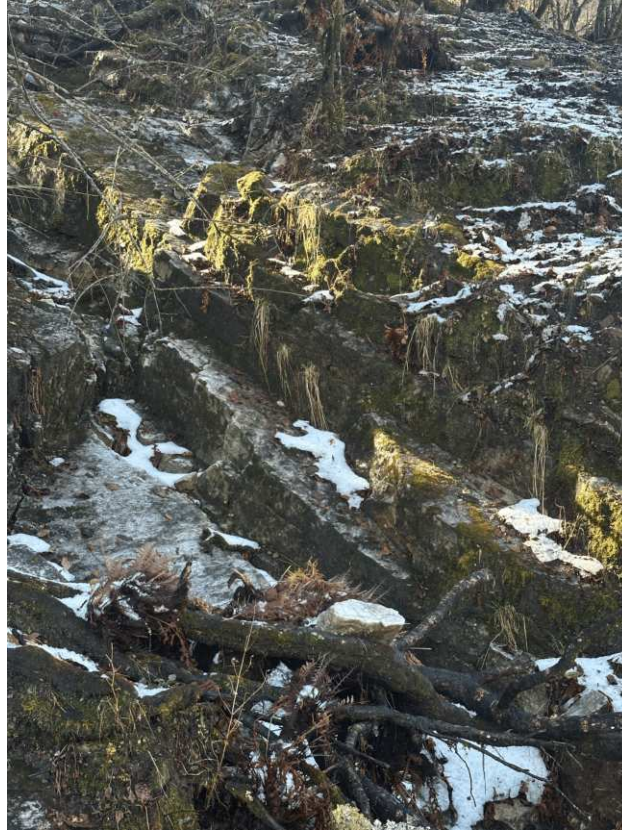


Figure 4: **Slope-parallel foliation planes exposed in a landcut along the Dayara Bugyal trail.** The dip direction of the foliation matches the downslope direction, creating the geometric condition under which pore water pressure can mobilize entire slabs along pre-existing planes of weakness.

Water converts the structural predisposition into actual movement. Phyllite with slope-parallel foliation is the standing cause, always present regardless of weather. Monsoon rainfall is the triggering cause. Infiltration along foliation planes raises pore water pressure, reducing effective normal stress and frictional resistance along surfaces that mineral structure has already weakened (Kanungo & Sharma, 2014). Simultaneously, absorbed water increases gravitational driving force by adding mass to the slope (Allen et al., 2015). When pore pressure rises to the point that the driving force exceeds the resisting force, a translational landslide occurs along a foliation plane. Phyllite on flat ground would not fail; granite on the same slope would be far more stable. Both standing and triggering causes are necessary, and together they are sufficient to cause landslides.

Stripped of soil and vegetation, exposed foliation surfaces undergo accelerated weathering, with sericite converting to weaker clay minerals such as kaolinite and vermiculite, further reducing shear strength (Silwal et al., 2024). Root cohesion, measured at 25 to 94 kPa in comparable forested slopes (Schmidt et al., 2001), disappears and takes decades to recover, thereby increasing the probability of the next failure.

Conclusion

The landslide susceptibility of the Garhwal Himalaya arises from interacting geological and environmental processes. Proterozoic shales metamorphosed into phyllite, whose foliation reflects sericite's weak basal cleavage expressed at the slope scale. Where this foliation aligns parallel to hillsides, it forms natural failure planes. Rainfall then increases pore-water pressure, reducing effective stress and triggering translational landslides along these planes. Subsequent loss of soil, vegetation, and root cohesion, combined with weathering of sericite into weaker clays, further destabilizes slopes. Each failure increases vulnerability to the next, linking mineral structure, tectonic history, and climate into a self-reinforcing hazard system.

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